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Force Control Using Internal Spring in Electrostatic Linear Motors and Switching Between Position and Force Control

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***Abstract*—This article proposes a method for controlling force and position in a synchronous direct-drive electrostatic linear motor. Using a spring-like behavior of synchronous motors, the proposed controller regulates the contact force in a manner similar to that of series elastic actuators. Discussions regarding similarities and differences between series elastic actuators and the proposed method imply that their dynamic behaviors are different. The proposed controller consists of a force control part and a position control part, one of which is automatically selected based on the operating conditions. The position and force controllers each independently command the velocity based on the position or force error. The lower of the two velocities is selected, allowing a smooth automatic transition between the two control modes. Based on the selected velocity, the driving signal is calculated under the assumption of synchronous operation and fed to the actuator. The proposed control method is simulated and experiments are performed using prototype electrostatic linear motors. The experimental results confirm the smooth transition between the position and force control modes, as well as the good controllability of the contact force.**

***Index Terms*—Direct-drive, electrostatic film motor, force control, robotics, series elastic actuator (SEA), synchronous motor.**

I. Introduction

VARIOUS types of robot arms, made for both industrial and household purposes, have become available in stores [1], [2], [3]. However, safety concerns arise when these robot arms operate near people, which hinders their adoption. Those arms need to approach objects and make contact for task execution, while ensuring safety in case of unexpected contact with a person or obstacle [4], [5], [6]. One effective solution to achieve this is to use direct-drive actuators [7], [8], [9], which do not involve fragile reduction gears or other devices in the drive force transmission path, reducing the risk of damage due to excessive impact force. As one of such direct-drive actuators, this work focuses on synchronous and light-weight electrostatic linear motors [10].

The electrostatic linear motors can directly generate tens to hundreds of newtons of force without a reduction gear [10]. In a report [11], an actuator weighing 235 g managed to generate 259 N. A recent study [12] demonstrated that a two-degree-of-freedom (DoF) robot arm driven by electrostatic linear motors could effectively lower impact forces. In the study, the step-out phenomenon of a synchronous electrostatic motor was utilized as a variable torque limiter to regulate the contact force. When the contact force reached a certain limit, the motor retreated by stepping out to reduce the force (generally, a synchronous motor's operation steps out when it is subjected to a large disturbance). While safe contact was ensured in the method proposed in Osada et al.'s [12] work, it is not enough to accomplish tasks; it is necessary to regulate contact force during contact with target objects, which remains as a challenge for this actuator.

To achieve such a function, a typical solution is using force feedback control [13], [14], [15], [16], in which actual force is measured by a force sensor and is fed back to a controller to minimize the error. In series elastic actuator (SEA) [17], a soft spring connected to the output of the actuator replaces a force sensor, which realizes straightforward force control through motor position control [18], [19], [20]. SEA uses a physical spring, which makes the robot arm larger and heavier and lowers the frequency response of the system [21], [22]. Therefore, this study introduces a novel concept of "internal spring" and explores the possibility of force control similar to SEA without using a physical spring. Since the proposed controller of this work performs both the force control and position control in the velocity dimension, the study proposes a smooth switching between position and force control using limiters on the velocity command, thus enabling trajectory tracking in free space and force regulation during contact in a unified framework.

The rest of this article is organized as follows. Section II introduces the structure and basic operating principles of electrostatic linear motors, followed by the presentation of the novel concept of the internal spring and a comparison with SEA. Section III

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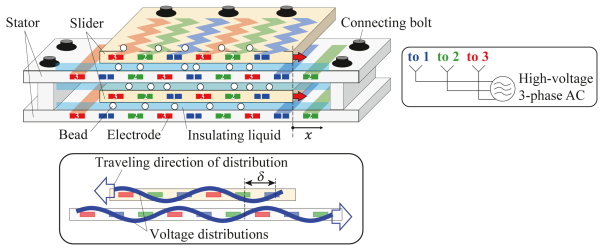


Fig. 1. Structure of electrostatic linear motor with multiple layers. The stator has straight electrodes, whereas the slider has skewed electrodes for force ripple reduction. The red arrows on the slider indicate the direction of movement. The electrode numbers indicate the phases of the applied voltage. The application of the three-phase voltage creates traveling voltage distributions as shown in the lower box.

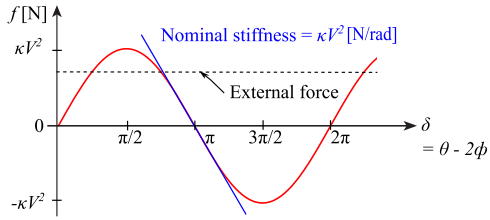


Fig. 2. Force characteristics of the electrostatic motor against the force angle δ (red line). During synchronous operation, the angle shifts within $\delta = [\pi/2, 3\pi/2]$, where the driving force changes almost linearly (blue line).

describes the proposed controller, analyzing the behaviors of position and force control modes and explaining the transition between them. Section IV verifies the controller proposed in Section III using simulations. Section V presents experimental results obtained using prototype one-DoF electrostatic linear motors, demonstrating successful mode transition. The section also investigates force control performance in both static contact and in sudden collisions. Finally, Section VI concludes this article.

II. Concept of Internal Spring

A. Internal Spring in Electrostatic Linear Motors

The electrostatic linear motor used in this work is made of flexible printed circuit films and operates within a dielectric liquid, as shown in Fig. 1. The motor consists of two sets of films, which are referred to as stator and slider. Each set consists of several films with the same design. The simplest structure utilizes only one pair of stator and slider films; for a larger output force, multiple pairs of films are stacked as shown in the figure.

In the motor, both the stator and slider films have three-phase electrodes with the same regular interval p, called the electrode pitch. For the prototype used in this work, p is 100 μm. To reduce force ripple, the stator electrodes have a straight shape, while the slider electrodes have a zigzag shape to introduce the skew effect [23]. Since the electrodes are three-phase electrodes, the same phase appears at multiples of three times the electrode pitch. This length $\lambda = 3p$ is referred to as the periodic length.

The motor is operated by applying three-phase sinusoidal high voltages to both the stator and slider. This creates traveling voltage distributions on both the stator and the slider. As the connection orders of the three-phase voltages are reversed in the stator and slider (see Fig. 1), the two voltage distributions travel in opposite directions. These two distributions generate an electrostatic force, and the slider will move to compensate for the speed difference in the two distributions, which is $6p$ times the frequency of the applied sine wave. This indicates that the motor is synchronous; the motor speed is proportional to the applied frequency, and the displacement of the motor is proportional to the phase of the applied sine wave.

The force equation of the motor is expressed as follows [23]:

$$f = \kappa V^2 \sin(\theta - 2\phi) = \kappa V^2 \sin(\delta) \tag{1}$$

where θ is the slider position expressed in terms of the electric angle, which is defined as $\theta = 2\pi x/\lambda$, where x is the actual position of the slider (see Fig. 1), and ϕ is the phase of the applied voltage. The force angle δ is defined as $\delta = \theta - 2\phi$. The δ is equivalent to the spatial phase difference between the two voltage distributions. The coefficient κ is the force coefficient and V is the amplitude of the applied sine voltage wave. The maximum voltage is limited by the electric breakdown strength of the motor components, which defines the maximum force of the motor.

Fig. 2 plots the force characteristics. During synchronous operation, the force angle δ shifts within $[\pi/2, 3\pi/2]$ to balance the external force. If the external force is constant, the angle δ $(= \theta - 2\phi)$ is also constant. Therefore, an increase in ϕ results in an increase in θ, which results in synchronous operation. When the external force changes, the angle δ shifts such that the resulting force balances the external force. From a kinematic perspective, if the force angle δ is regarded as a length, this behavior is equivalent to a spring of a length δ balancing an external force. In other words, in the synchronous state, the motor behaves like a spring whose natural length is π. Therefore, this characteristic of the motor is termed as an "internal spring" in this study.

Although this spring is nonlinear, it is modeled as a linear spring with a nominal stiffness of K_s, which is derived as

$$K_s = -\left.\frac{df}{d\delta}\right|_{\delta=\pi} = \kappa V^2 \tag{2}$$

for the analyses in this work. The control method proposed in this work is based on this internal spring.

B. Comparison With SEA

In this work, the motor's thrust force is controlled by the length of the internal spring, as described in the next section. Although such a control scheme looks quite similar to that for SEA, the different locations of the springs result in different dynamic behaviors, which are discussed in the following.

Fig. 3(a) shows a simple model of an SEA, in which a spring (K) connects a motor mass (M_m) and a load (M_l) in series. The internal viscosity of the motor (D_m) and the viscosity of the load (D_l) are also considered. This model is represented by the following equations:

$$M_{\mathrm{m}}\ddot{x}_{\mathrm{m}} = f_{\mathrm{m}} + K(x_{\mathrm{l}} - x_{\mathrm{m}}) - D_{\mathrm{m}}\dot{x}_{\mathrm{m}} \tag{3}$$

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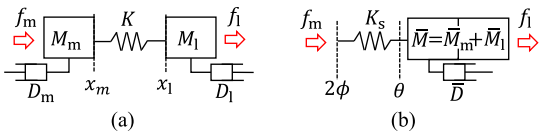


Fig. 3. Model comparison between SEA and the force control using the internal spring. In SEA, the model has two inertia and one elasticity, whereas the system using the internal spring has only one inertia and one elasticity. (a) SEA. (b) Proposed.

$$M_l\ddot{x}_l = f_l - K(x_l - x_m) - D_l\dot{x}_l \tag{4}$$

where x_m and x_l are the positions of the motor and load, respectively. Force f_m represents the force of the motor and f_l is the force exerted by the load or contact force.

On the other hand, if the internal spring is used, the spring is located before the mass of the slider. Therefore, the situation is modeled, as in Fig. 3(b). By using electric angle to represent the displacement, the following equations are obtained

$$0 = f_m + (\theta - 2\phi)K_s \tag{5}$$

$$\bar{M}\ddot{\theta} = f_l - (\theta - 2\phi)K_s - \bar{D}\dot{\theta} \tag{6}$$

where the motor force in (1) is replaced by a linear spring force, $\bar{M} = \lambda M/(2\pi)$ is the equivalent mass and $\bar{D} = \lambda D/(2\pi)$ is the equivalent damping coefficient in the electric angle representation. Since there is no spring between the motor output and the load, their positions coincide; that is, $x = x_m = x_l = \lambda\theta/(2\pi)$.

These two models differ in the number of masses connected to the springs, which affects their behaviors. In SEA, if the motor is moving in free space (i.e., $f_l = 0$), it behaves as a two-inertia system, which complicates the system behavior and makes control difficult. When the motor is in contact with an obstacle (i.e., x_l is constant), the system reduces to a one-inertia system and the force can be controlled through the change in x_m.

On the other hand, when the internal spring is used, the system is a one-inertia system during motions in free space. In the force control during contact with a static object (i.e., θ is constant), no mass motion is involved and f_l can be controlled by the phase ϕ without any dynamics. In both situations, the system using the internal spring involves fewer masses. This means that the use of the internal spring leads to simpler dynamics, making control easier.

III. Proposed Position–Force Controller

A. Concept and Overview of Proposed Controller

This work proposes a controller for synchronous electrostatic motors, that controls position when the motor (or the arm driven by the motor) is moving in free space and controls force when the motor (or arm) is in contact with an object or the environment. Such control could also be realized in a way similar to field orientation control, in which the force angle δ is kept at $\pi/2$ by selecting ϕ to satisfy $\phi = (\theta - \pi/2)/2$. Then, the instantaneous force can be controlled by changing the voltage amplitude, as $f = \kappa V^2$. Position can also be controlled by building a position feedback loop. This control strategy is quite common for electromagnetic motors and has also been implemented for electrostatic linear motors [24]. However, it requires the high-precision detection of the relative electrode positions of the stator and slider. For example, if 5% error is permitted for force control, the detection error for the electrode positions must be less than 0.05 rad in electric angle (if the linear spring behavior is assumed for the motor), which is equivalent to 2.4 μm for a motor with $p = 100$ μm. If an external position sensor is used to estimate the relative electrode position, which is internal information of the motor, this precision is not easy to achieve, as the electrode films used in the motor are flexible. To address this problem, built-in sensors were proposed for direct detection of the electrode position [24], [25], [26]. However, implementing those built-in sensors requires a rather complicated setup or signal processing. In the control method of this work, the use of such built-in sensors is not necessary, although they can also be used if applicable.

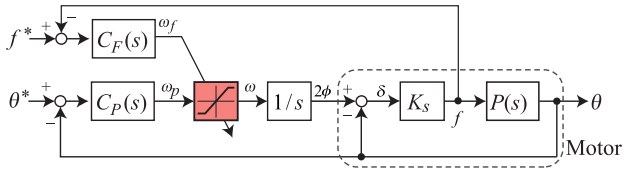


Fig. 4. Position–force switching controller. The limiter shown by the red block controls the transition between force and position control. The set value of the limiter is determined by ω_f.

Fig. 4 shows the proposed controller. Variables f^* and θ^* represent force and position references, respectively. The controller consists of a position control loop and a force control loop, which are integrated using a limiter for the commanded velocity ($\omega = d\theta/dt$).

In the proposed controller, when the motor (and arm) is moving in free space, a position control loop is automatically selected. On the other hand, when the motor is in contact with an obstacle, the force control loop is selected. This automatic selection is done by the limiter block. The limiter is utilized here to prevent instability that may arise from control switching based on force threshold judgments. Although not explicitly shown in the figure, the voltage amplitude is kept constant during operation, which differentiates this controller from field orientation control.

The proposed integrated controller was devised since, in the concept of this work, the motor force is controlled via displacement (the length of the internal spring), which seems to work well with position control. However, as the target variables are different in force control and in position control, they cannot be simply integrated in the dimension of displacement (or angle). Therefore, this work integrates the two systems by differentiating each variable into the dimension of angular velocity. The behavior of each controller is explained and analyzed in the following.

B. Position Control in Free Space

The lower blocks in Fig. 4 show the position control part of the proposed controller. This position controller is essentially the

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same as that utilized in the previous study [12]. The controller requires a position sensor to detect the slider position. However, unlike the displacement sensor for the field orientation control, this position sensor does not have to be very accurate (required accuracy depends on applications) and can be compactly placed outside the motor.

In this controller, the motor is driven in synchronous operation, where the force angle $\delta = \theta - 2\phi$ is in the synchronous range of $[\pi/2, 3\pi/2]$. As the motor displacement is almost proportional to the phase of the applied sine wave in synchronous operation, the main objective of the controller is to determine an appropriate phase ϕ for following a target trajectory. To achieve this, the controller first calculates the velocity command ω based on the position error.

If we assume that δ is kept almost constant, the following relationship holds for ω and ϕ:

$$\omega = \frac{d\theta}{dt} = \frac{d}{dt}(\delta + 2\phi) \approx 2\frac{d\phi}{dt}. \tag{7}$$

Therefore, integrating the velocity command ω, yields twice of target phase, 2ϕ. Using ϕ, three-phase sine waves are generated with a preset voltage amplitude and then applied to the motor. More specifically, the applied voltage set is

$$\mathbf{V} = V[\sin(\phi), \sin(\phi - 2\pi/3), \sin(\phi + 2\pi/3)] \tag{8}$$

where V is the amplitude. Then, the motor generates force, as expressed in (1), which is represented by the internal spring stiffness K_s in the block diagram for simplicity.

The transfer function of the position control loop is

$$\frac{\Theta(s)}{\Theta^*(s)} = \frac{K_\mathrm{s}P(s)C_\mathrm{P}(s)}{s(1 + K_\mathrm{s}P(s)) + K_\mathrm{s}P(s)C_\mathrm{P}(s)} \tag{9}$$

where $\Theta(s)$ and $\Theta^*(s)$ are the Laplace transforms of θ and θ^*, respectively. If we assume simple dynamics for the motor (and arm), such as

$$P(s) = \frac{1}{Ms^2 + Ds} \tag{10}$$

the transfer function can be rewritten as

$$\frac{\Theta(s)}{\Theta^*(s)} = \frac{K_sC_\mathrm{P}(s)}{K_sC_\mathrm{P}(s) + s\,(Ms^2 + Ds + K_s)} \tag{11}$$

where M is the equivalent mass of the motor slider (and arm) and D represents the internal damping. For simplicity, this analysis adopts the simplest proportional controller, $C_\mathrm{P}(s) = K_\mathrm{p}$. The condition for stability is obtained from the Routh stability criterion as $(D/M) > K_p$. The controller is type-I, with one pole at the origin, and thus there will be no steady-state error for a step input.

C. Force Control During Contact

The upper block in Fig. 4 shows the force control part of the proposed controller. This controller achieves force control in a manner similar to that of SEA. As shown in the diagram, the force controller controls the phase ϕ. If a motor (and arm) is in contact with an obstacle, such as a wall, the slider position θ remains constant. Therefore, the change in ϕ is directly reflected in the change in the force angle δ, which means that the force controller indirectly controls the force angle δ. As δ represents the deformation of the internal spring, the concept of the proposed force controller is similar to that of SEA, where force is controlled through spring deformation.

In the proposed controller, the output force is measured or estimated and fed back to the controller. The controller outputs a velocity command based on the force error, which is then integrated to obtain 2ϕ. Then, the voltage set is determined based on (8).

For the simplest proportional controller, $C_\mathrm{F}(s) = K_\mathrm{f}$, the transfer function of the force control loop is given as

$$\frac{F(s)}{F^*(s)} = \frac{K_\mathrm{s}C_\mathrm{F}(s)}{s(1 + K_\mathrm{s}P(s)) + K_\mathrm{s}C_\mathrm{F}(s)} \tag{12}$$

where $F(s)$ and $F^*(s)$ are the Laplace transforms of f and f^*, respectively. With the simple assumption for $P(s)$, as done in (10), we have

$$\frac{F(s)}{F^*(s)} = \frac{K_\mathrm{f}K_\mathrm{s}(Ms + D)}{Ms^2 + (D + MK_\mathrm{f}K_\mathrm{s})s + (K_\mathrm{f}K_\mathrm{s}D + K_\mathrm{s})}. \tag{13}$$

The system is always stable as it is second-order with positive coefficients in the denominator. When the motor is in contact with a rigid wall, which means that the motor position θ is fixed, the transfer function reduces to

$$\frac{F(s)}{F^*(s)} = \frac{K_\mathrm{f}K_\mathrm{s}}{s + K_\mathrm{f}K_\mathrm{s}}. \tag{14}$$

This is a first-order lag system with a dc gain of 1 and a time constant of $1/(K_\mathrm{f}K_\mathrm{s})$. The force can thus be controlled without steady-state error.

If a derivative action (with a gain of K_fd) is added, which is the case in the experiment described later, the transfer function changes to the following form:

$$\frac{F(s)}{F^*(s)} = \frac{K_\mathrm{f}K_\mathrm{s}}{(1 + K_\mathrm{fd}K_\mathrm{s})s + K_\mathrm{f}K_\mathrm{s}} \tag{15}$$

where the derivative action is applied only to the fed-back signal (not to the reference input), realizing proportional–derivative (P-D) control. If $K_\mathrm{fd}K_\mathrm{s}$ is sufficiently large, the function reduces to

$$\frac{F(s)}{F^*(s)} \approx \frac{K_\mathrm{f}}{K_\mathrm{fd}s + K_\mathrm{f}} \tag{16}$$

which is a first-order lag system with a time constant, $K_\mathrm{fd}/K_\mathrm{f}$.

D. Position–Force Control Switching

In the proposed controller, shown in Fig. 4, an automatic transition between the position control (in free space) and force control (during contact) is achieved using the limiter for the velocity command ω. The limiter has a variable limit, which is determined by the commanded velocity from the force controller, ω_f. When the commanded velocity from the position controller, ω_p, is smaller than the limit (ω_f), ω_p passes through the limiter block, which means that the controller is in position control mode. When ω_p exceeds the limit (ω_f), the output of the

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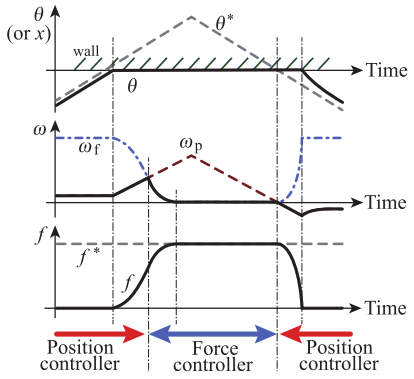


Fig. 5. Behavior of the position–force control switching controller. The velocity commands ω_p and ω_f are determined based on the position and force error, respectively. When ω_p exceeds ω_f, ω_f is adopted as the velocity command, which realizes a transition from position control to force control.

limiter becomes equal to ω_f, which means that the controller is in force control mode.

The expected behavior is shown in Fig. 5. First, the motor is in free space and is given a triangular trajectory θ^* that penetrates a wall (top plot in the figure). The force controller is given a constant force reference, which results in an initial constant velocity command ω_f. As the motor makes contact with the wall, the position controller tries to move the motor further by increasing ω_p. This action contracts the internal spring and the contact force increases, which results in a decrease in the force error. As a result, the velocity command from the force controller, ω_f, is decreased and eventually becomes smaller than ω_p. After ω_p exceeds ω_f, the limiter is activated and its output becomes equal to ω_f, which means that the system is in force control mode. Then, as the reference trajectory retreats, the position error decreases and eventually becomes negative. As a result, the output of the position controller ω_p becomes lower than the limit and the system transitions to position control mode again.

In this switching, the behavior of the limiter is critical. The behavior can be rigorously formulated as follows. When the output of the force feedback controller, ω_f, is positive, the limiter should operate as

$$\omega = \begin{cases} \omega_\mathrm{p} & (\omega_\mathrm{p} \le \omega_\mathrm{f}) \\ \omega_\mathrm{f} & (\omega_\mathrm{p} > \omega_\mathrm{f}). \end{cases} \tag{17}$$

For negative ω_f, the inequality signs should be reversed.

IV. Simulation

A. Switching Between Position Control and Force Control

To verify the proposed position–force control switching, a simulation was conducted in MATLAB/Simulink. As contact with a perfectly rigid body is unrealistic, a wall with unidirectional viscoelastic properties was assumed as the obstacle. The wall was arranged at $x = 0$ that generated a reaction force in the range of $x > 0$. In the simulation, the controller shown in Fig. 4 was implemented. The motor force was calculated using (1).

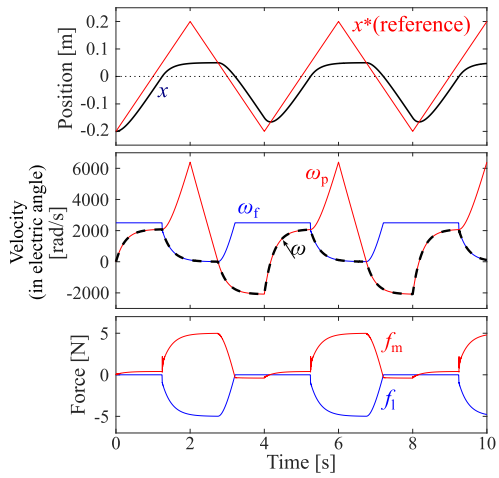


Fig. 6. Simulation results for pushing a wall with the position–force controller. In the range of $x > 0$, the stiffness and damping of the wall were set as 100 N/m and 5 N · s/m, respectively. The contact force f_l in the bottom plot is taken in the opposite direction for clarity. ($V = 1$ kV, $\kappa = 5 \times 10^{-6}$ N/V^2, $\lambda = 3 \times 10^{-4}$ m, $K_\mathrm{p} = 4$ s^{-1}, $K_\mathrm{f} = 1000$ rad/s · N, $M = 0.05$ kg, $D = 2$ N · s/m, $f^* = 5$ N).

The motor was supposed to have a position sensor to monitor the slider displacement, given by $x = 3p\theta/2\pi$, as well as a force sensor to monitor the contact force on the obstacle. Both the position and force controllers adopted simple proportional controllers.

Two different methods were tested for force measurement to realize force feedback. One method uses the force sensor that directly monitors the contact force, which is referred to as f_l. The other method estimates the motor output force from (1), which is referred to as f_m, using the position sensor. When the motor is stationary, these two forces are equivalent; during motion, they are different due to inertia and damping.

Fig. 6 shows the simulation results. The motor was commanded to move at a constant speed toward a wall with a finite stiffness. After contact, it was commanded to retreat at the same absolute speed. The same motion was then repeated. For this plot, the contact force f_l was fed back to the force feedback controller. A similar result (not shown) was obtained when the motor force f_m was fed back.

The bottom plot clearly shows that the contact force was controlled to match the reference force of 5 N during contact. The middle plot confirms that the control mode was automatically changed via the selection of the smaller value between ω_f and ω_p. During motion in free space, the position controller output was adopted, whereas during most of the contact time, the force controller output was adopted. This confirms that the force and position control, as well as switching between them, can be successfully achieved without oscillation.

B. Response to Applied Disturbance

For the two force measurement methods, completely different responses were obtained when a large position error occurred. In Fig. 7, the motor initially kept a constant position. Then, an impulsive force that resembles a sudden collision was applied.

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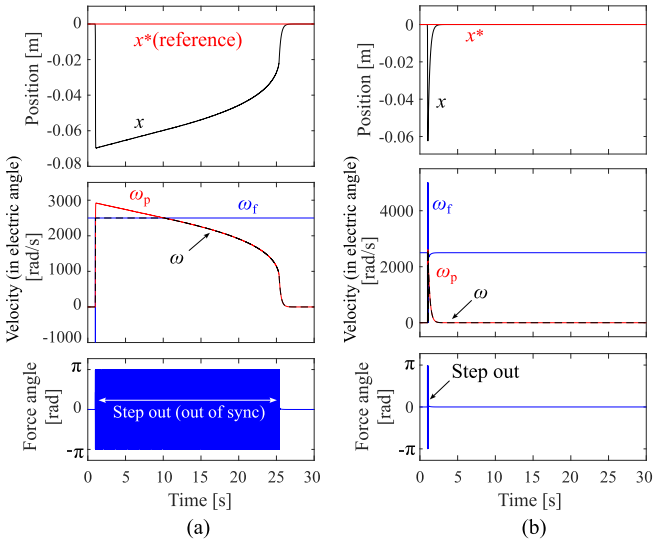


Fig. 7. Simulation results for sudden collision. An impulsive force (a force for 0.1 s at 25 N) was applied at 1 s. ($V = 1$ kV, $\kappa = 5 \times 10^{-6}$ N/V^2, $\lambda = 3 \times 10^{-4}$ m, $K_\mathrm{p} = 4$ s^{-1}, $K_\mathrm{f} = 1000$ rad/s$\cdot$ N, $\kappa V^2 = 5$ N, $M = 0.05$ kg, $D = 2$ N $\cdot$ s/m, $f^* = 5$ N). (a) Contact force f_l fed back. (b) Motor force f_m fed back.

The impulsive force (25 N) was much larger than the motor's maximum force (5 N). The motor stepped out as soon as the impulsive force was applied at 1 s. With this quick step-out motion, the motor successfully mitigated a sudden collision. The motor then recovered and went back to its original position. This recovery motion was different between the two force measurement methods.

When the force sensor measurement, f_l, was used as in Fig. 7(a), the motor was commanded to have an excessively fast recovery speed due to the large position error. However, the motor could not respond due to its inertia and synchronized motion could not be achieved. On the other hand, when the estimated motor output force, f_m, was used, step-out during recovery was suppressed. For f_l, the sensor simply measures the contact force, which becomes zero in free space. However, even in free space, inertia, and internal damping are loaded onto the motor. For f_m, the effects of inertia and damping can be taken into account to prevent step-out.

In summary, the results suggest the following. If the force is measured by a force sensor installed at the end effector, the contact force can be precisely measured. However, during movement in free space, such a force sensor does not measure the internal loss and can thus cause continual step-out. To suppress such step-out, f^* or K_f can be set to smaller values to limit the maximum speed. This, however, requires fine adjustment of f^* or K_f through trial and error.

On the other hand, if the force is estimated from the force angle, step-out can be suppressed, since the generating force of the motor is always monitored. If the generating force approaches the reference value of the force controller that is set under the maximum force of the motor at the given voltage amplitude (κV^2), the controller switches to force control mode before step-out occurs. However, the estimated force in this method is different from the actual contact force due to the inertia of the slider mass and damping. Therefore, this method can introduce some errors in contact force control. The errors are negligible in the steady state and thus the contact force can be correctly regulated if the contact is stationary. If, however, there is a motion during contact, there can be a small error in the force regulation. Such cases include contacts with movable objects or elastic walls and following a curved surface with keeping a constant contact force. Performance in such situations should be investigated in future work.

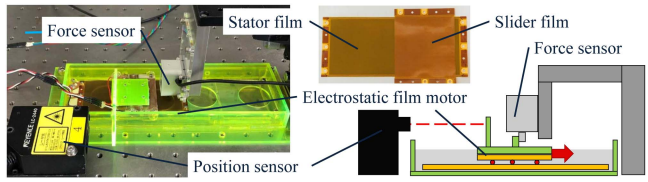


Fig. 8. Single-layer setup for force control experiment (left and right) and films used in setup (middle). A force sensor was installed on the wall.

In practice, the force estimation method is difficult to implement. The displacement θ in (1) is the relative displacement of the electrodes of the stator and slider, which is internal information. As discussed earlier, this internal information is difficult to estimate from the displacement measured from the outside since the stator and slider are made of flexible films. The deformation of the films or other elastic components can easily create estimation error on the order of tens of micrometers, which is significant compared to the electrode pitch (100 μm in this work). To implement this method, built-in displacement sensors [25] that can directly monitor the internal information must be utilized, but the implementation is rather complicated (and therefore was avoided in the experiments of this work).

If both the force sensor and the built-in displacement sensor are available, combining both informations can eliminate the above-mentioned errors. The force sensor can be used for contact force regulation, and the built-in sensor can be used to prevent step-out.

V. Experiment

A. Force Step Response

The position control of this work is essentially the same as that in Osada et al.'s [12] work, and its performance has already been investigated. Therefore in this work, first, force control was tested with a low-inertia prototype. A single-layer electrostatic linear motor was assembled, as shown in Fig. 8. A pair of flexible printed circuit films, which functioned as the stator and slider, were arranged in a plastic housing that was filled with dielectric liquid (Fluorinert FC-770, 3 M) to prevent discharges in the surrounding air. The effective actuation area (i.e., the overlapping area between the stator and slider) was 25 cm^2 (5 cm $\times$ 5 cm). A small number of glass beads with a diameter of 20 μm were scattered between the films to reduce friction and maintain an appropriate gap.

The electrode pitch p was 100 μm and the slider electrodes were skewed to achieve smooth force characteristics [23]. The

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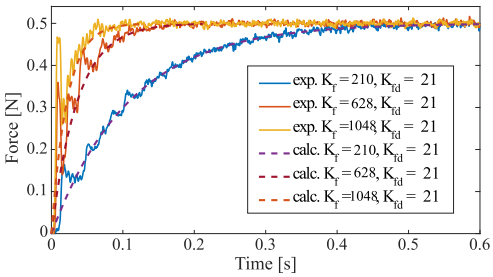


Fig. 9. Step responses in force control experiment for three feedback gain values. The dashed lines represent the theoretical responses.

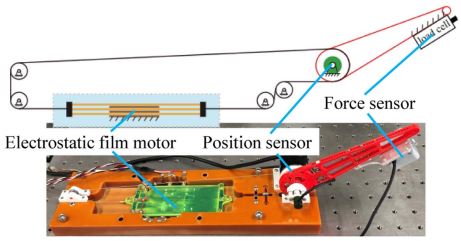


Fig. 10. One-DoF arm driven by an electrostatic motor using 11 pairs of films. A position and force sensor were installed on the arm for feedback control.

stator film was fixed to the housing, whereas the slider film was connected to a movable plastic plate. The total moving mass M was 30 g. For force calibration, a force sensor (Kyowa, LVS-2KA) was fixed on a wall so that the thrust force of the motor could be measured. From measurement using the sensor, the coefficient κ was estimated as 1.1×10^{-6} N/V^2.

In the following force control experiment, the voltage amplitude V was fixed at 1000 V. As the internal stiffness K_s is equal to κV^2, the stiffness of the internal spring K_s is 1.1 N/rad when the displacement is measured in terms of the electric angle; this value is equivalent to 22 kN/m. For the force control experiment, the position control loop in Fig. 4 was disabled and only the force control loop was implemented. The measured force (f_l) was used for feedback control. Although the simulations in Section IV-B described two types of force signals, f_l and f_m, the use of f_m requires a built-in sensor that is rather complicated. Therefore, the experiment used f_l, which was obtained from the force sensor, for force control.

First, force control in a static situation was tested. The slider was arranged such that the thrust force could be measured by the force sensor on the wall. Then, 0.5 N was set as the force reference. Fig. 9 shows the results. The experiment was conducted with three different gains, namely $K_f = 210$, 628, and 1048 rad/(s · N), which are equivalent to 10, 30, and 50 mm/(s · N) respectively. In addition, a derivative action was added with a gain of $K_{fd} = 21$ rad/N or 1 mm/N, since the response was too oscillatory without it. The derivative gain was applied to the feedback signal (θ), not to the error ($\theta^* - \theta$).

In Fig. 9, although some oscillation appears, the outputs are roughly first-order lag curves that correspond to the theoretical estimations. These results confirm the validity of the theoretical estimation regarding force control. It was also confirmed that the proposed control method does not show any steady-state error in contact force control.

B. Position–Force Control Switching

A simple one-DoF arm was fabricated to test position–force control switching. The prototype is shown in Fig. 10. The arm was rotated by an electrostatic motor via steel wires. The motor consisted of 22 films, which are comprised of 11 stator films and 11 slider films, to increase the output. The designs of the films are identical to those in Fig. 8. The motion of the arm between 0° (upright) and 90° (right) corresponded to a motor displacement of about 30 mm. An encoder (EXTCOM, EV10S20000X) was attached to the pulley, to measure the angle of the arm. A force sensor (Kyowa, LVS-2KA) was installed at the end of the arm (140 mm from the axis). As the pulley that was driven by a steel wire had a radius of 20 mm, the motor force was reduced to one-seventh at the force sensor. The measured force was directly fed back to the force controller. As the force sensor directly measured the contact force, the friction of the pulleys or other losses did not affect the force control result.

The position–force control switching in Fig. 4 was implemented using proportional controllers. The voltage amplitude was fixed at 1000 V and the maximum force κV^2 was experimentally determined as 16 N.

First, the transition from position control to force control was tested. Ramp trajectories with speeds of 29 and 86 deg/s were set as the position reference such that the arm hit an obstacle arranged on the floor, as shown in Fig. 11(a). For the force controller, the reference force was set to 1 N (which corresponds to a motor force of 7 N, given the 1:7 force ratio).

The results of collisions at two different speeds are shown in Fig. 11(b). The horizontal axis was adjusted such that the collisions occur at $t = 2.8$ s. For either reference speed, the arm speed reached the target speed after an initial transient response. After contact, as shown in Fig. 11(b), the force converged to 1 N without a large overshoot for both cases. The fluctuation in contact force is substantial, making it difficult to discern the time constant, yet the transition is completed within 30 ms. This suggests that the position control was smoothly replaced by force control.

For testing transitions in both directions (position to force and force to position), like the simulations in Figs. 5 and 6, the arm was commanded to track a sinusoidal trajectory (0.5 Hz). An obstacle positioned in the middle of the trajectory (at 45° of the arm angle), as depicted in Fig. 11(a) was utilized to test the transitions. During this test, a target force of 0.5 N was used.

The results of the contact during sinusoidal wave tracking are depicted in Fig. 12. Although the transition from position to force control shows a large impulsive overshoot due to a collision against a hard target, the transition was realized without any other oscillations. For the transition from force back to position control, there was no overshoot because the contact force was released upon termination of contact, enabling smooth transition.

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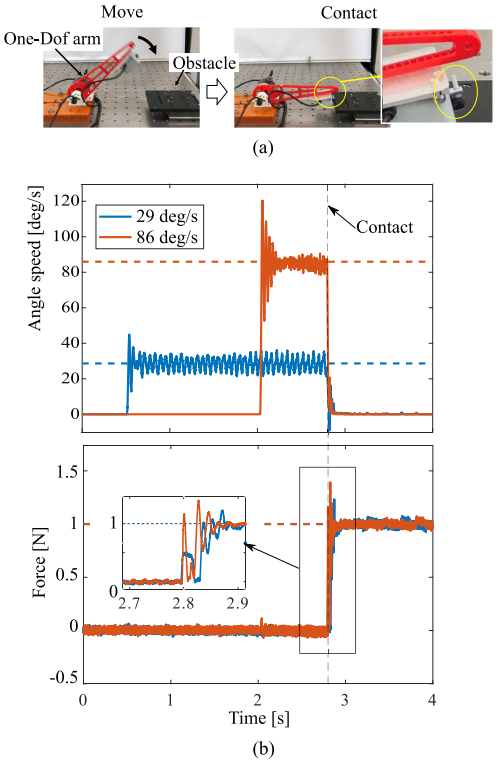


Fig. 11. Results of position–force control switching. The arm was first operated in position control mode with a ramp trajectory (either 29 or 86 deg/s). Then, at 2.8 s, the arm hit an obstacle. After the collision, the contact force was maintained at the reference force of 1 N. The feedback gains were $K_{\mathrm{p}} = 20$ s^{-1} and $K_{\mathrm{f}} = 628$ rad/N· s. (a) Photographs showing motion during experiments. (b) Experimental angular speed (top) and force (bottom).

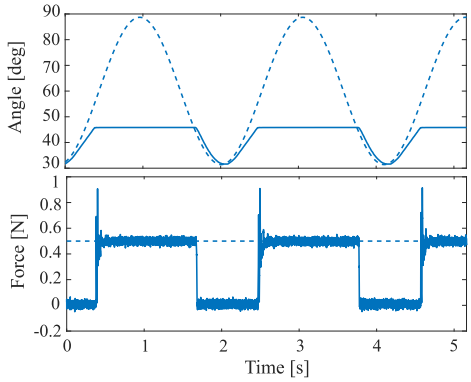


Fig. 12. Results of position–force control switching during tracking of a sinusoidal trajectory. In the beginning, the arm was operated in position control mode. Then, the arm hit an obstacle during the motion (at 45°). After the collision, the contact force was regulated to 0.5 N, which matched the given reference force. The feedback gains were $K_{\mathrm{p}} = 20$ s^{-1} and $K_{\mathrm{f}} = 628$ rad/N· s.

VI. Conclusion

This study proposed to utilize an internal spring in a synchronous direct-drive electrostatic motor to enable the robot arm to approach and make contact with an object with any desired force. It was shown that position and force can be controlled in the same velocity dimension, allowing for smooth switching. The motion in free space is controlled using synchronous operation. The contact force is controlled in a manner similar to that of SEA, in which the contraction of the internal spring is controlled. The two controllers individually command the motor velocity. The lower speed among the two commanded ones is selected.

The experimental results confirmed that the internal spring of a synchronous motor can be utilized for force control, eliminating the need for a physical spring. The results also demonstrated that the proposed control method can transition from position control to force control within 30 ms without oscillation and achieve the commanded contact force with no deviation during contact.

Although the behavior during this transition showed some variations, such as an impulsive overshoot, this would be mitigated by adjusting the gains and the threshold, which should be investigated in future work. In addition, exploring stability under extreme conditions remains a potential issue for future works. The experiment only showcased the results of the one-DoF arm making contact with a flat rigid body. Our future work will address higher DoF motions, as well as contact against curved surfaces and deformable objects.

In this work, the proposed control method was implemented for an electrostatic motor, but it should be also valid for other types of direct-drive synchronous motors, since the internal spring exists in the other synchronous motors, as well. Future work will address the implementation of the proposed method for other motors to confirm its validity for a wide range of synchronous motors.

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